Choosing Your Dataset

Students practice making a variety of chart types and then begin to investigate a real world dataset, which they will continue to work with for the remainder of the course.

Lesson Goals	Students will be able to… • Explain why they chose their dataset • Describe their dataset • Make subsets from their dataset
Student-facing Lesson Goals	• Let's all choose an interesting dataset to investigate.
Prerequisites	• Simple Data Types • Contracts • Bar and Pie Charts
Materials	• Lesson Slides • Animals Starter File • Blank Dataset Starter File for Bootstrap:Data Science • Global Food Supply & Production • Dataset Exploration • Consider and Analyze (Page 19) • My Dataset (Page 20) • Data Cycle: Shape of My Dataset (Page 21)

Preparation	• Decide how much choice you're ready to offer your students before you begin. Research shows that choice increases student engagement! But focusing the whole class on a single dataset is also an option.
	  ○ Would focusing your students on a single dataset make this doable for you? Because you teach younger students who might need more scaffolding? Or because you are new to teaching data science and managing fewer moving parts would increase your confidence? We recommend focusing on Global Food Supply & Production.
	  ○ Are you ready to jump straight into supporting your students in working on a wide range of topics of their choosing? We have a full dataset library!

Glossary

categorical data :: data whose values are qualities that are not subject to the laws of arithmetic

data science :: the science of collecting, organizing, and drawing general conclusions from data, with the help of computers

dataset :: a collection of related information that is composed of separate elements, but can be manipulated as a unit by a computer

quantitative data :: number values for which arithmetic makes sense

random sample :: a subset of individuals chosen from a larger set, such that each individual has the same probability of being chosen

statistical inference :: using information from a sample to draw conclusions about the larger population from which the sample was taken

Review *20 minutes*

Overview

Students practice making lots of chart types, focusing specifically on the "Consider Data" step in the Data Cycle and how it can be used alongside Contracts to help go from questions to code.

Launch

Let's get some practice isolating the Rows and Columns needed to answer various questions, and use our knowledge of Contracts to help turn those questions into working code!

Complete Consider and Analyze.

Be sure to review student answers.

Choosing a Dataset *30 minutes*

Overview

Students select a dataset that interests them, and do some thinking about *why* it interests them, *what questions they'd like to answer* and *what hypotheses they have* . They'll be analyzing this data for a long time, so it's critical to ensure a high degree of buy-in before signing off on a student's choice!

If you are opting to focus your whole class on a single dataset, we recommend skipping to the Exploring Your Dataset section of this lesson and using the dataset provided there. *(It focuses on global food supply and production through environmental / geographic / cultural lenses and the variables were carefully selected to make sure it lends itself well for all kinds of data displays and discussions. You can, of course, opt to choose any dataset you'd like, from our library or otherwise.)*

Launch

Data Science: it's all about YOU!

What data matters to ***you?*** What questions do ***you*** care about? We live in a world filled with data, gathered about almost every subject you can imagine.

- Climate sensors are gathering data on temperature, humidity, oxygen and more…practically everywhere on the globe.
- Census data tracks the number of different groups of people, as well as their education, income level, and more.
- Companies like Facebook, Amazon, and Google gather massive amounts of data on the websites you visit, what you chat about online, what you purchase, etc.

This data is used to set public policy, draw voting districts, approve drugs, calculate school funding, decide which advertisements you see, and more.

- Where else do you see data being gathered?
- What are some other ways data is used in the world around you?

Below is a list of every ***dataset*** already provided to students, with a corresponding Starter File that instantly imports the (cleaned) data into Pyret.

Students can also find their own dataset , and use this Blank Dataset Starter File for Bootstrap:Data Science. See this tutorial video for help importing your own data into Pyret.

Investigate

Have students choose a dataset that is interesting to them and save a copy of it in their programs!

Looking for a shorter list? We've starred a few good beginner datasets.

The Environment & Health

Global Waste by Country 2019	[Dataset Starter File]
World Cities' Proximity to the Ocean	[Dataset Starter File]
Earthquakes	[Dataset Starter File]
Air Quality, Pollution Sources & Health in the U.S.	[Dataset Starter File]
Health by U.S. County	[Dataset Starter File]
COVID in the U.S. by County	[Dataset Starter File]
Arctic Sea Ice	[Dataset Starter File]

Politics

Countries of the World	[Dataset Starter File]

Gerry Mandering	[Dataset Starter File]
Marijuana Laws & Arrests by State 2018	[Dataset Starter File]
LAPD Arrests 2010-2019	[Dataset Starter File]
NYPD Stop, Search & Frisk 2019	[Dataset Starter File]
Refugees 2018	[Dataset Starter File]
State Demographics	[Dataset Starter File]
U.S. Income	[Dataset Starter File]
U.S. Jobs	[Dataset Starter File]
U.S. Voter Turnout 2016	[Dataset Starter File]

Sports

Esports Earnings	[Dataset Starter File]
MLB Hitting Stats	[Dataset Starter File]
NBA Players	[Dataset Starter File]
NFL Passing	[Dataset Starter File]
NFL Rushing	[Dataset Starter File]

Entertainment

★Movies	[Dataset Starter File]
IGN video game Reviews	[Dataset Starter File]
International Exhibition of Modern Art	[Dataset Starter File]
North American Pipe Organs	[Dataset Starter File]
Pokemon	[Dataset Starter File]
Music	[Dataset Starter File]

Education

College Majors	[Dataset Starter File]
U.S. Colleges 2019-2020	[Dataset Starter File]
★R.I. Schools	[Dataset Starter File]
Evolution of College Admissions in California	[Dataset Starter File]

Nutrition

Soda, Coffee & Other Drinks	[Dataset Starter File]
Fast Food Nutrition	[Dataset Starter File]

Would you like to contribute a dataset of your own, or is there something you'd like to change about one of ours?

Synthesize

Have students share which datasets they chose, and why they are interesting or important to them. What questions did they come up with?

For the rest of this course, you'll be learning new programming and ***Data Science*** skills, practicing them with the Animals Dataset and then applying them to you own data.

Exploring Your Dataset *Start Today… continue in Upcoming Lessons*

Overview

Students apply what they've learned about describing and making subsets from the Animals Dataset to their own dataset. *If your students will all be focusing on the same dataset, we recommend using Global Food Supply & Production.*

Launch

By now you've already learned what to do when you approach a new dataset.

- With the Animals Dataset, you first read the data itself, and wrote down your Notices and Wonders.
- You described the columns in the Animals Dataset, identifying which were ***categorical*** and which were ***quantitative***, and whether they were Numbers, Strings, Booleans, etc.
- You took ***random samples*** of the dataset, to explore ***inference*** and probability.

Now, you're doing to do the same thing *with your own dataset.*

Investigate

- Look at the spreadsheet or table for your dataset. What do you **Notice**? What do you **Wonder**?
- Complete My Dataset, making sure to include at least two questions that *can* be answered by your dataset and one that *cannot* .
- Save a copy of your starter file. In the Definitions Area, use `random-rows` to define **at least three** tables of different sizes: `tiny-sample` , `small-sample` , and `medium-sample` .

Today we will begin working on the Dataset Exploration, which will prepare students for writing their research papers. We will return to this in upcoming lessons. We are just going to work on the first section for now.

- Make a copy of Dataset Exploration, and open the starter file for your dataset.
- Complete the first set of questions in the exploration paper.
- What are the categorical columns in *your* dataset? How are those values distributed?
- Turn to Complete Data Cycle: Shape of My Dataset, and use the Data Cycle to generate pie and bar charts.
- What do these charts tell you? Add the images of these charts - along with your interpretation! - to the "Making Displays" section of the exploration document.
- Do these displays bring up any interesting questions? If so, add them to the end of the document.

Synthesize

Have students share their findings. Were any of them surprising?